

Progress of PREFAB: Preference based Self-Annotation on a Low-Budget

JaeYoung Moon

Gwangju Institute of Science and Technology
Gwangju, Republic of Korea
super_moon@gm.gist.ac.kr

Georgios N. Yannakakis*

University of Malta
Msida, Malta
georgios.yannakakis@um.edu.mt

YouJin Choi

Gwangju Institute of Science and Technology
Gwangju, Republic of Korea
chldbwls@gm.gist.ac.kr

Kyung-Joong Kim*

Gwangju Institute of Science and Technology
Gwangju, Republic of Korea
kjkim@gist.ac.kr

ABSTRACT

Self-annotation is significantly important in affective computing field but has the limitation of heavily relying on human cognition. To address this, we propose a method named PREFAB: Preference-based self-Annotation on a low-Budget (PREFAB). This paper shares our progress, focusing on predicting player arousal levels from game trajectories using a RankNet-based preference learning approach. Experimental results demonstrate that our method significantly outperforms existing benchmarks in terms of training performance. In terms of practicality, it successfully captures the tendency of the changes in players' arousal better than traditional approach.

CCS CONCEPTS

• **Human-centered computing** → **Interaction techniques**; • **Computing methodologies** → *Cognitive science*; **Knowledge representation and reasoning**.

KEYWORDS

Affective Computing, Preference Learning, Self-Annotation, Player Modeling

ACM Reference Format:

JaeYoung Moon, YouJin Choi, Georgios N. Yannakakis, and Kyung-Joong Kim. 2024. Progress of PREFAB: Preference based Self-Annotation on a Low-Budget. In *Companion of the 2024 ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp Companion '24)*, October 5–9, 2024, Melbourne, VIC, Australia. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3675094.3678379>

1 INTRODUCTION

In the field of affective computing, labeling data required for user modeling primarily represents the internal state of humans, such as emotion, immersion, enjoyment, and arousal [4, 5, 7, 13, 14]. Due to the lack of objective measurement methods for human internal

states, self-annotation, where users directly annotate their own experiences, is commonly used [5, 7, 14]. However, self-annotation has several limitations because it relies heavily on human cognition.

First, self-annotation can be inaccurate as the number of steps in the sequence of the annotation task increases or time gap between the task and the annotation grows larger. Secondly, individuals have different criteria, so even when facing the same situation, people may feel different experiences, and even if they feel the same type of experience, the degree can vary greatly. Finally, self-annotation is time-consuming. Generally, when performing active tasks such as gaming, driving, or controlling something, real-time self-annotation is barely possible, necessitating post-annotation. In such cases, the annotation process may require as much time as the main task. As annotators become fatigued from these long process, errors in the annotations are likely to increase.

To address these issues, we propose *PREFAB: Preference based self-Annotation on a low-Budget (PREFAB)*: a method to improve the efficiency of self-annotation in the gaming domain by detecting peak intervals where players' expected arousal levels change drastically via preference learning, asking players to annotate these intervals, and then interpolating the remaining intervals based on the annotated peak intervals.

This paper shares an intermediate stage of the research process for the proposed method, focusing on arousal prediction via preference learning, which is a part of the described process above.

2 BACKGROUND

2.1 (Self) Annotation Research

In the field of affective computing, various interfaces have been proposed to annotate the user's actual experience as consistently as possible such as PAGAN [7], DARMA [5], and RCEA [14]. These interfaces aimed to provide users with a more intuitive way to annotate their own state. However, they still required annotation work for the entire sequences, which did not overcome the limitation of the heavy reliance on human cognition.

Meanwhile, effort to reduce annotation costs have not been unique to affective computing. In many fields that require large amounts of data, such as classification, segmentation, and regression, extensive research on low-budget annotation has been conducted. Various methods have been employed, including domain adaptation [11], active learning [10], and semi-supervised learning

*Co-corresponding Authors



This work is licensed under a Creative Commons Attribution International 4.0 License.

[3]. However, all of these problems predicted cardinal or numeric values, not relative (ordinal) values such as human's emotion and experience.

Herein, we propose a method to reduce the self-annotation cost in affective computing via preference learning, covering ordinal data.

2.2 Ordinal Nature and Preference Learning

Yannakakis *et al.* emphasized the ordinal nature of the human cognition in affective computing [13]. Ordinal nature refers to the relative order or ranking of data points rather than their absolute values. In deep learning, ordinal nature has often been implemented by the RankNet architecture [1]. In this study, we constructed the PREFAB network based on the RankNet structure including the Transformer's encoder architecture[12].

$$\begin{aligned} \Phi(x_i) = p_i, \Phi(x_j) = p_j, p_j - p_i = p_{ij}, \\ P(y_{ij} = 1) = P(p_j > p_i) = \sigma(p_{ij}) = \frac{1}{(1 + e^{-p_{ij}})} \end{aligned} \quad (1)$$

$$BCE(x_i, x_j) = -y_{ij} * \log(\sigma(p_{ij})) - (1 - y_{ij}) * \log(1 - \sigma(p_{ij})) \quad (2)$$

RankNet (Φ) is a neural network architecture designed for learning to rank (relative and ordinal). It takes a pair of inputs x_i and x_j and trains to predict the relative order p_{ij} between them. Where x is an input data, i and j indicate each input segment, and p_{ij} is ordinal relationship between input segments. The probability that p_j is greater than p_i (which equal to the label $y_{ij} = 1$) is calculated using the sigmoid function σ (Equation 1). Basically, the RankNet model handles binary classification (BC) problem (greater or not). Therefore, the network is trained using a binary cross-entropy loss function that measures the difference between the predicted probability of the order and the true probability of the order (Equation 2). Extending this concept, we aimed to predict three conditions: greater ($k = 2$), equal ($k = 1$), and less ($k = 0$). We employed ordinal cross-entropy (OCE) loss [9] to train the three-class RankNet based on the probability in Equation 3. Where, c_0 and c_1 indicate cutoff of each class for the OCE.

$$P(y_{ij} = k) = \begin{cases} \sigma(c_0 - p_{ij}), & \text{if } k = 0, p_i > p_j \\ \sigma(c_1 - p_{ij}) - \sigma(c_0 - p_{ij}), & \text{if } k = 1, p_i = p_j \\ 1 - \sigma(c_1 - p_{ij}), & \text{if } k = 2, p_i < p_j \end{cases} \quad (3)$$

3 RESEARCH OVERVIEW AND CURRENT MILESTONE

In this study, we propose a novel approach to predict player state (e.g. arousal in this work) in a game sequence using a preference learning-based model. The primary objective is to reduce the annotation burden on players while maintaining the accuracy of the self-annotation of the entire sequences.

The overall process is as follows. First, a RankNet-based arousal prediction model is trained to estimate the arousal levels experienced by a player during gameplay. Next, peak regions where the arousal levels exhibit significant changes are identified. These peak regions are considered to be the most informative and emotionally salient moments in the game sequence. Subsequently, we present

only the identified peak regions to the player for self-annotation, thereby markedly reducing the quantity of data that requires manual labeling. Finally, we interpolate the remaining regions to estimate the unlabeled arousal levels.

Currently, we have completed the initial stage of the proposed approach, which involves predicting arousal levels using the RankNet-based model. This milestone lays the foundation for the subsequent steps of peak region identification, selective annotation, and interpolation.

4 MODEL ARCHITECTURE

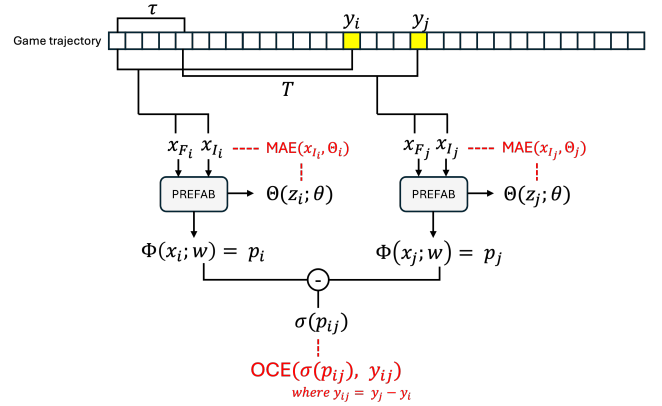


Figure 1: Overall Process to Train PREFAB

4.1 Input and Output

Figure 1 demonstrates the training process of the PREFAB. The input data consists of T-frames of images (X_{I_i}) and game features (X_{F_i}) for the i -th segment (sequence). For a given segment from n to $n+T$ frames, assuming the segment starts at the n -th frame, the annotated arousal level at the $(n+T)$ -th step is assumed to represent the entire T-frames of the segment. Pairs of segments with a τ -frames interval are labeled as to whether the later sequence is greater than, lower than, or equal to the former sequence by subtracting y_{n+T} from $y_{n+T+\tau}$. The data pairs for the input and the label are sampled within the same game trajectory.

4.2 Network Architecture and Loss Function

Figure 2 shows an architecture of the PREFAB module. The proposed network takes images $\{x_{I_i}\}_{n+T}^{n+T+\tau}$ and game features $\{x_{F_j}\}_n^{n+T}$ of the i -th segment as input and predicts the relative value p_i . Images are compressed into a latent space using an autoencoder consisting of several convolutional neural network (CNN) layers to extract valuable features. The extracted latent vector is concatenated with the game features and further processed for feature extraction using fully-connected (FC) layers. The extracted features are then fed into a transformer network. Finally, the transformer output is reduced to a 1-dimensional value by FC layers.

As shown in the Figure 1, the RankNet architecture follows a Siamese network structure, where two segments $\{x_i\}_n^{n+T}$ and

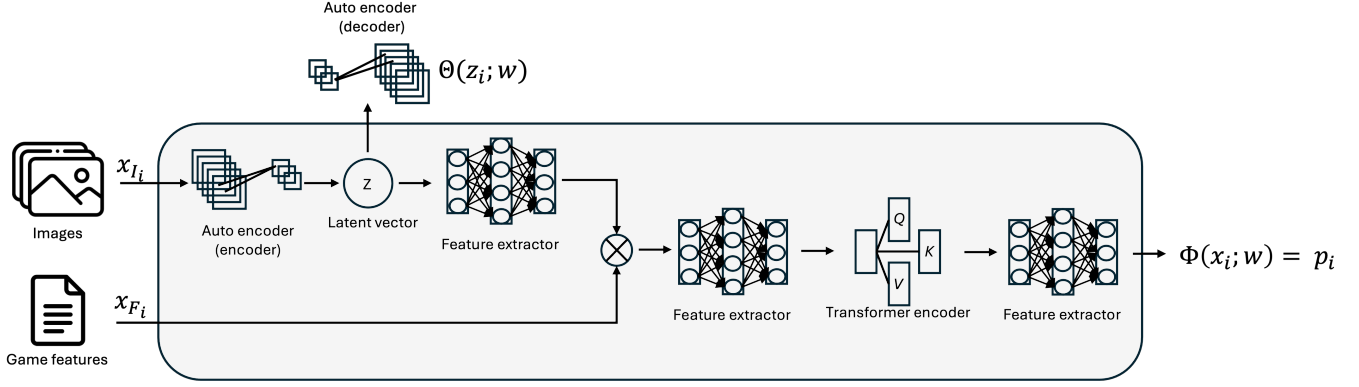


Figure 2: Architecture of PREFAB Module

$\{x_j\}_{n+\tau}^{n+\tau+T}$ are processed independently, sharing the PREFAB module $(\Phi(\cdot; w))$ to obtain their respective values $(\{p_i\}_n^{n+T}$ and $\{p_j\}_{n+\tau}^{n+\tau+T})$. The difference between $\{p_i\}_n^{n+T}$ and $\{p_j\}_{n+\tau}^{n+\tau+T}$ is fed into a sigmoid function $\sigma(\cdot)$. Then, the OCE loss (Equation 3) is computed to minimize the difference between $\sigma(p_{ij})$ and y_{ij} . In addition to the OCE loss, the autoencoder should be trained by comparing the output of the reconstructed images $\Theta(z; \theta)$ with the input images x_I . The mean absolute error (MAE) loss is employed for the autoencoder loss. Finally, the PREFAB model is trained by combining the OCE loss and the MAE loss. In Particular, the autoencoder converges very quickly to the optimum, and its loss is relatively larger than the OCE loss. Therefore, a scaling factor α is applied to balance the training (Equation 4).

$$L = OCE(\sigma(y_{ij}, p_{ij})) + \alpha * (MAE(x_{I_i}, \Phi(z_i; \phi)) + MAE(x_{I_j}, \Phi(z_j; \phi))) \quad (4)$$

5 EXPERIMENT

5.1 Dataset

We employed the AGAIN dataset [8] for the experiment. The AGAIN dataset is a collection of player experience annotations (arousal) for video game sessions, aimed at facilitating research in affective computing and player modeling in the context of video games. It includes game logs and videos for 9 games and players' self-annotated arousal levels for each game session. Each game session lasted a total of 2 minutes, and the game logs were recorded at a frame rate of 4. Since the AGAIN dataset is provided for learning ordinal nature, the self-annotated arousal levels were normalized between 0 and 1. There are more than a hundred game features in the dataset, including common game features across the 9 games and unique features for each game. Herein, we employed the common features and the unique features for the games used in the experiment. Due to the nature of arousal in games, there were many upward-sloping graphs, resulting in a ratio of less, equal, and greater labels of approximately 2:2:6. Therefore, we used oversampling and oversampling methods to balance the data.

In this work, we used only two games, Heist! and TopDown, which are shooter genre games. In the literature, the authors provided a simple benchmark solving a binary classification problem

using the AGAIN dataset through a Random Forest, and the model's prediction performances were the best with these two games.

5.2 Implementation Details

We followed the best practices of the prior works [2, 6, 8] for segmenting the input data: a 3-second window with a 1-second stride (gap) between segments. Since the AGAIN dataset was recorded at a frame rate of 4, 12 frames were fed into the PREFAB model with 4-frame interval.

The experiments were conducted using 4 GPUs from an Nvidia DGX A100 system. The input image size was resized to (360, 240) pixels. The model was trained for 60 epochs using the Adam optimizer with an initial learning rate of 0.001 and a batch size of 24. The cutpoints for the ordinal classification were set to $c_0 = -1$ and $c_1 = 1$, and the autoencoder loss weight α was 0.001.

The dataset was split into train, validation, and test set with a 7:2:1 ratio. During the training phase, the train and validation sets were used. Input segments were sampled from individual game trajectories, but the order of the sampled pairs was randomly shuffled. For the test phase, the test set consisted of n (depending on the data size) non-shuffled game trajectories. Each game trajectory was divided into 3-second segments, which were then fed into the PREFAB model to obtain predicted arousal levels for the entire game.

6 RESULT

6.1 Preference Learning Performance

We validated the preference learning result $(\sigma(p_{ij}))$ on the validation set as a 3-class classification problem, categorizing the output into less, equal, or greater classes using cutpoints of -1 and 1. We compared the classification performance of the PREFAB with the results reported in the previous work [8].

Table 1 shows the result. The binary classification accuracy in the AGAIN benchmark reached about 80% for the game Heist! and 82.5% for the game TopDown. Our approach outperformed the benchmark, achieving 95.8% accuracy for both games, even though it covered the 3-class classification problem. This result reflects how well the preference learning-based PREFAB model captured the ordinal nature of the players' arousal tendencies.

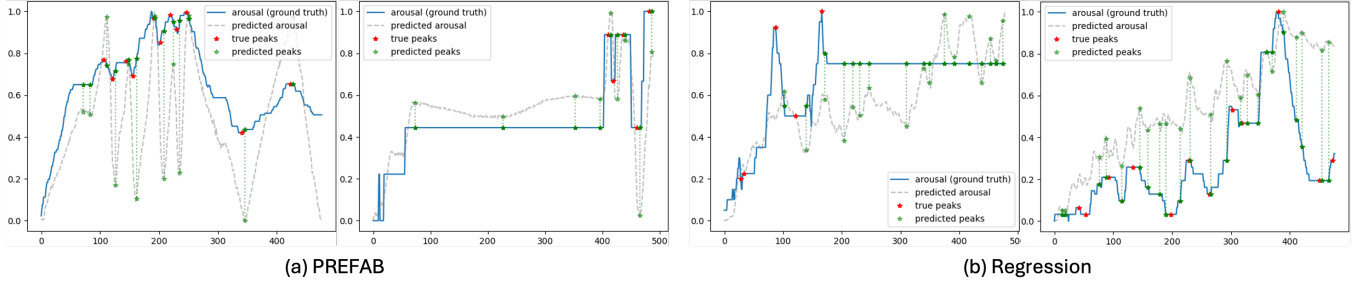


Figure 3: Peak Region Comparison between Predicted Arousal Graph and Ground Truth Arousal Graph.

Table 1: Classification Performance.

Approach	Game	
	Heist!	TopDown
AGAIN (2-class)	under 80%	82.5%
PREFAB (3-class)	95.8%	95.8%

6.2 Peak Region Comparison

We compared the peak regions from the predicted arousal graphs by the PREFAB model for unseen games with the peak regions of the ground truth (GT), which are the self-annotated arousal levels in the AGAIN dataset. By comparing the peaks of the predicted arousal graph with the peaks of the actual self-annotated arousal graph, we examined how well the PREFAB model represents the changes in the players' arousal. To evaluate the superiority of our approach, we compared it with a network that follows the same architecture as the PREFAB but is trained using a typical regression learning method.

Table 2: Overlap Percentage between the Peaks from the Predicted Arousal Graph and from the Ground Truth Arousal Graph.

Approach	Game	
	Heist!	TopDown
Regression	26.1%	33.8%
PREFAB	64.9%	69.5%

Figure 3 demonstrates the predicted arousal graphs (gray dotted line) and GT arousal graphs (blue line) with peaks of each graph (green for the predicted arousal and red for the GT). The predicted arousal graphs were rescaled between 0 and 1 for easy comparison to the GT. This shows that the PREFAB outperformed the conventional regression-based approach in predicting player arousal levels from game trajectory. A qualitative analysis of the predicted arousal graphs reveals that the PREFAB better captures the changing tendency of the GT arousal peaks regardless scales. Notably, in regions where the player's arousal remained stable, the PREFAB model showed a similar gradual change, whereas the regression-based model showed extreme fluctuations and detected

multiple peaks, which is inconsistent with the GT. Furthermore, a quantitative evaluation was conducted by calculating the probability of the predicted arousal peaks p overlapping with the nearest GT peaks G within a specified distance D (Equation 5). Table 2 showed that the peaks detected in the regression-based arousal graphs only overlapped with the GT peaks in 26.1% and 33.8% for the Heist! and TopDown games, respectively. In contrast, the peaks detected in the PREFAB-based arousal graphs overlapped with the GT peaks in 64.9% and 69.5% for the same games, more than doubling the overlap ratio.

$$\text{Ratio} = \frac{1}{n} \sum_{i=1}^n I(p_i) \quad (5)$$

$$\text{where, } I(p_i) = \begin{cases} 1, & \text{if } \min_{g_j \in G} |p_i - g_j| \leq D/2 \\ 0, & \text{otherwise} \end{cases}$$

These findings suggest that the proposed approach was not only twice as accurate as the conventional method but also has the potential to provide more efficient self-annotation sampling points.

7 FUTURE WORK

This work represents an early stage of the PREFAB study. Based on this work, several future studies can be pursued. First, we plan to further refine the algorithm used to identify peak regions where arousal changes rapidly. A more sophisticated peak detection method could potentially be developed to optimize the trade-off between providing effective annotation points to the user and reducing the annotation cost. Second, we aim to develop a robust interpolation method to estimate arousal levels for the remaining region as accurately as possible based on the annotated peak regions. This will involve exploring various interpolation techniques and evaluating their performance in the context of arousal prediction. Finally, we intend to conduct user studies to validate our approach empirically, involving a diverse set of games. These studies will provide valuable insights into the effectiveness of the PREFAB in real-world scenarios and help identify areas for further improvement. By addressing these aspects, we strive to advance the field of affective computing in games, robotics, and other diverse areas, ultimately paving the way for more engaging and emotionally resonant experiences tailored to individual users.

ACKNOWLEDGMENTS

This research was supported by Culture, Sports and Tourism R&D Program through the Korea Creative Content Agency grant funded by the Ministry of Culture, Sports and Tourism in 2022 (Project Name: Development of artificial intelligence-based game simulation technology to support online game content production, Project Number: R2022020070).

REFERENCES

- [1] Chris Burges, Tal Shaked, Erin Renshaw, Ari Lazier, Matt Deeds, Nicole Hamilton, and Greg Hullender. 2005. Learning to rank using gradient descent. In *Proceedings of the 22nd international conference on Machine learning* (Bonn, Germany) (ICML '05). Association for Computing Machinery, New York, NY, USA, 89–96. <https://doi.org/10.1145/1102351.1102363>
- [2] Elizabeth Camilleri, Georgios N Yannakakis, and Antonios Liapis. 2017. Towards general models of player affect. In *2017 Seventh International Conference on Affective Computing and Intelligent Interaction (ACII)*. IEEE, 333–339.
- [3] Zhenghao Chen, Rui Zhang, Gang Zhang, Zhenhuan Ma, and Tao Lei. 2020. Digging into pseudo label: a low-budget approach for semi-supervised semantic segmentation. *IEEE Access* 8 (2020), 41830–41837.
- [4] Youjin Choi, JooYeong Kim, and Jin-Hyuk Hong. 2022. Immersion measurement in watching videos using eye-tracking data. *IEEE Transactions on Affective Computing* 13, 4 (2022), 1759–1770.
- [5] Jeffrey M Girard and Aidan G C. Wright. 2018. DARMA: Software for dual axis rating and media annotation. *Behavior research methods* 50 (2018), 902–909.
- [6] Phil Lopes, Georgios N Yannakakis, and Antonios Liapis. 2017. Ranktrace: Relative and unbounded affect annotation. In *2017 Seventh International Conference on Affective Computing and Intelligent Interaction (ACII)*. IEEE, 158–163.
- [7] David Melhart, Antonios Liapis, and Georgios N Yannakakis. 2019. PAGAN: Video affect annotation made easy. In *2019 8th international conference on affective computing and intelligent interaction (ACII)*. IEEE, 130–136.
- [8] David Melhart, Antonios Liapis, and Georgios N Yannakakis. 2022. The arousal video game annotation (AGAIN) dataset. *IEEE Transactions on Affective Computing* 13, 4 (2022), 2171–2184.
- [9] Fabian Pedregosa, Francis Bach, and Alexandre Gramfort. 2017. On the Consistency of Ordinal Regression Methods. *Journal of Machine Learning Research* 18, 55 (2017), 1–35. <http://jmlr.org/papers/v18/15-495.html>
- [10] Pengzhen Ren, Yun Xiao, Xiaojun Chang, Po-Yao Huang, Zhihui Li, Brij B Gupta, Xiaojiang Chen, and Xin Wang. 2021. A survey of deep active learning. *ACM computing surveys (CSUR)* 54, 9 (2021), 1–40.
- [11] Cristiano Saltori, Paolo Rota, Nicu Sebe, and Jurandy Almeida. 2022. Low-budget label query through domain alignment enforcement. *Computer Vision and Image Understanding* 222 (2022), 103485.
- [12] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. 2017. Attention is all you need. *Advances in neural information processing systems* 30 (2017).
- [13] Georgios N Yannakakis, Roddy Cowie, and Carlos Busso. 2018. The ordinal nature of emotions: An emerging approach. *IEEE Transactions on Affective Computing* 12, 1 (2018), 16–35.
- [14] Tianyi Zhang, Abdallah El Ali, Chen Wang, Alan Hanjalic, and Pablo Cesar. 2020. Rcea: Real-time, continuous emotion annotation for collecting precise mobile video ground truth labels. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*. 1–15.